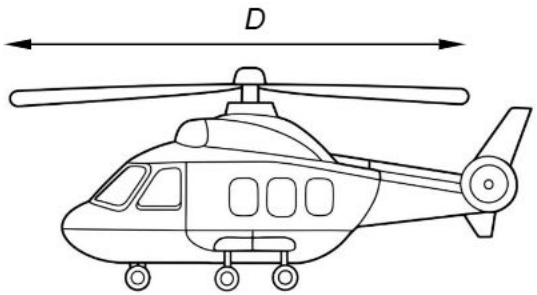
maximum downward velocity.

6 A helicopter hovers stationary in mid-air. Its rotor blades sweep out a circular path of diameter D and push down a continuous column of air of density ρ at a uniform speed v. The air above the blades is taken to be stationary.

Which expression correctly represents the average power P delivered to the air column by the blades?



A $\frac{1}{8}\pi\rho D^2v^3$ **B** $\frac{1}{2}\pi\rho D^2v^3$ **C** $\frac{1}{4}\pi\rho D^2v^3$ **D** $\frac{1}{4}\pi\rho D^2v^2$

Ans: **A**

Mass flow rate: $A = \pi r^2 = \frac{\pi D^2}{4}$

$$\frac{\Delta m}{\Delta t} = \rho A v = \rho\left(\frac{\pi D^2}{4}\right)v \Rightarrow F = \frac{\Delta m}{\Delta t}v = \frac{1}{4}\rho\pi D^2v^2 \quad \text{(not needed)}$$

$$P = \frac{\text{change in KE}}{\text{time taken}} = \frac{1}{2}\left(\frac{\Delta m}{\Delta t}\right)v^2 = \frac{1}{2}\left[\rho\left(\frac{\pi D^2}{4}\right)v\right]v^2$$

$$= \frac{1}{8}\pi\rho D^2v^3$$

B: forgot to divide diameter

C: Used $P = Fv$ directly. (forgot factor of ½ in KE formula)

D: Used $P = F$